

9	(a) energy required to separate the nucleons (in a nucleus) or energy required to separate the protons and neutrons in a nucleus (or energy released when nucleons combine (to form a nucleus)/energy released when protons and neutrons combine to form a nucleus)	M1	
	either completely or to infinity (either free protons and neutrons or from infinity)	A1	[2]
	(b) (i) either different forms of same element or nuclei having same number of protons with different numbers of neutrons	M1	
		A1	[2]
	(ii) 1784 MeV (accept min. 3 s.f.) 7.57 MeV	A1	
		A1	[2]
	(c) (i) $\lambda = \ln 2 / (7.1 \times 10^8 \times 365 \times 24 \times 3600) = 3.1 \times 10^{-17} \text{ s}^{-1}$	B1	[1]
	(ii) $A = \lambda N$ $5000 = 3.1 \times 10^{-17} \times N$ $N = 1.61 \times 10^{20}$	C1	
	mass = $235 \times (1.61 \times 10^{20}) / (6.02 \times 10^{23})$ = 0.063 g (accept min. 2 s.f.)	C1	
		A1	[3]

Page 6	Mark Scheme	Syllabus	Paper
	Cambridge International AS/A Level – October/November 2015	9702	43

Section B